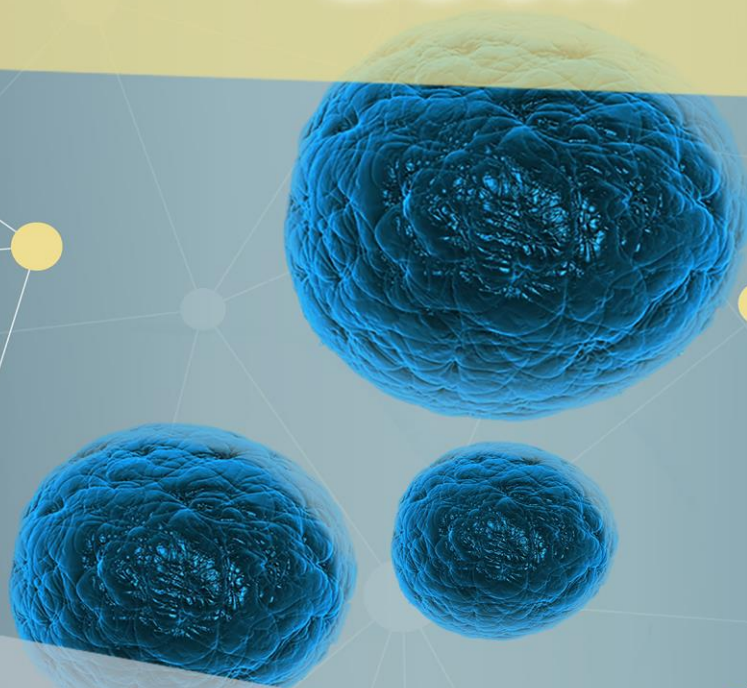


Serbian Association for Cancer Research

**5<sup>th</sup> CONGRESS OF SDIR:  
TRANSLATIONAL POTENTIAL OF  
CANCER RESEARCH IN SERBIA**

**ABSTRACT  
BOOK**



**Virtual event  
December 3**

**2021**

**EACR  
sponsored**

**THE FIFTH CONGRESS OF THE SERBIAN ASSOCIATION FOR CANCER RESEARCH**

with international participation  
"TRANSLATIONAL POTENTIAL OF CANCER RESEARCH IN  
SERBIA "

December 3, 2021, Virtual event  
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President of SDIR-5 Congress  
*dr sc. med. Mirjana Branković-Magić*

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## Molecular mechanisms of nanoparticle-mediated biological effects in doxorubicin treated cells

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**Background:** Engineered nanomaterials are at the leading edge of the rapidly advancing domain of nanomedicine, particularly in cancer research. Cytoprotective properties of nanoparticles due to their ROS scavenging capacity may significantly increase the therapeutic potential of cytotoxic drugs. Therefore, the main aim of our *in vitro* study was to investigate the expression of genes in fullereneol (FNM) treated cells in order to evaluate its therapeutic potential. **Material and methods:** Expression of genes involved in key cellular functions and processes, such as proliferation, apoptosis, redox regulation, DNA damage and repair was measured in K562 cells by the means of quantitative real-time PCR. RNA for cDNA synthesis was isolated from doxorubicin (DOX) treated erythroleukemia K562 cells, from FNM and from FNM+DOX treated cells. Gene expression was analyzed using comparative Ct method, and statistical analysis was performed using one-way Anova and Tuckey's post-hoc test. **Results:** Expression analysis of genes involved in antioxidative cell defense showed that FNM significantly suppresses DOX-induced inhibition of *MnSOD*, *GR* and *gGCS*. DOX-induced block in proliferation, as shown by decreased Ki67 expression was maintained also in FNM+DOX group of cells. FNM alone induced down-regulation of pro-apoptotic *BAX*, which is mediated by FNM-induced over-expression of BAX-inhibitor. Up-regulated BAX-inhibitor diminishes ER-stress-induced ROS accumulation and it is involved in induction of *HMOX*, *gGCS* and *GST* in K562 cells. Expression of *hOGG1* coding for base-excision repair enzyme was inhibited in FNM+DOX group, indicating that fullereneol contributes to increased K562 sensitivity to DOX. **Conclusion:** Fullereneol modulates redox status of DOX-treated K562 cells, it synergistically contributes to the proliferation block induced by DOX and exerts important ROS scavenging and apoptosis-modulating properties in erythroleukemia cells.

Key words: oxidative stress, nanoparticle, doxorubicin, apoptosis.

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